Recovering the Risk-Neutral Density Across Maturities: A Probabilistic and Computational Application of the Breeden–Litzenberger Result

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ABSTRACT

This paper presents a measure-theoretic and computational framework for recovering the risk-neutral density of an underlying asset across maturities via the Breeden–Litzenberger result. Starting from the probabilistic foundations of pricing under the risk-neutral measure $\mathbb{Q}$, we construct a pipeline that transforms raw option quotes into a smooth and arbitrage-free representation of market expectations. Market prices are first converted into discrete implied volatilities, which are then smoothed through SVI calibration to obtain an arbitrage-free implied-volatility surface. This surface is used to reconstruct the continuous call-price function via the Black–Scholes model, whose second-order finite-difference derivative—properly normalized—yields the risk-neutral density. The methodology is validated on synthetic data with known closed-form densities, showing high reconstruction accuracy. Finally, the full procedure is applied to S&P 500 option data.

1. Probabilistic Theory

Given a probability space $(\Omega, \mathcal{A}, \mathbb{P})$, we can define a random variable $X : \Omega \longrightarrow \mathbb{R}$. This induces a probability law P^X on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

If X is absolutely continuous with respect to Lebesgue measure, then it admits a density f_X such that:

$$P^X(A) = \int_A f_X(x)\, dx \qquad \forall A \in \mathcal{B}(\mathbb{R})$$

Moreover, if $h : \mathbb{R} \longrightarrow \mathbb{R}$ is a Borel function, then the composition $h(X)$ has a well-defined expectation[1] and the computation reduces to:

$$\mathbb{E}[\, h(X)\,] = \int_{\mathbb{R}} h(x) f_X(x)\, dx \tag{1}$$

2. Application to the Financial Setting

As far as the field of application is concerned, the probabilistic structure remains the same as in the theoretical case: we consider a probability space $(\Omega, \mathcal{A}, \mathbb{Q})$, and the random variable $S_T : \Omega \longrightarrow \mathbb{R}$, which describes the price of the underlying asset at maturity T. If it is continuous, it induces a probability law Q^{S_T} under $\mathbb{Q}$, and therefore a density $f^{\mathbb{Q}}_{S_T}$.

It is not possible to rely on the physical measure $\mathbb{P}$, since option prices do not reflect the real-world probability of the future evolution of the underlying, but instead the ***risk–neutral*** probability: $\mathbb{Q}$ is therefore an artificial measure under which the expected return of any traded asset must coincide with the risk–free rate r. As a consequence, the expected future price of the underlying grows at this rate, according to the continuous–compounding factor e^{rT}:

$$\mathbb{E}_{\mathbb{Q}}[\, S_T\,] = S_0\, e^{rT}.$$

Here, r denotes the risk–free rate, while e^{rT} represents the corresponding growth factor over the time interval $[0, T]$.

Therefore $\mathbb{Q}$ does not describe the actual evolution of the market, but it allows derivative prices to be expressed as the present value of the expected payoff, ensuring the absence of arbitrage.

In the absence of arbitrage, indeed, every derivative must have a price equal to the present value of the expectation of its payoff under the measure $\mathbb{Q}$.

It follows that the risk–neutral density $f^{\mathbb{Q}}_{S_T}$ represents the unique density consistent with the option prices observed in the market.

A European call option with strike K has payoff

$$\text{Payoff} = \max(S_T - K,\, 0) =: (S_T - K)^+$$

and theoretical price

$$\begin{aligned} C(K,T) &= e^{-rT}\, \mathbb{E}_{\mathbb{Q}}[\,\text{Payoff}\,] \\ &= e^{-rT} \int_{\mathbb{R}} (s-K)^+ f^{\mathbb{Q}}_{S_T}(s)\, ds \end{aligned}$$

This is theoretically justified by the fact that $S_T \in L^1(\mathbb{Q})$, and therefore also Payoff $\in L^1(\mathbb{Q})$. For the computation we set $h(s) = (s-K)^+$, which is a Borel function, and therefore we may use formula (1).

One can observe that the function $(S_T - K)^+$ is not K–differentiable in the classical sense, but it is differentiable in

[1] Formally, one should require $h \in L^1(P^X)$; however, for our purposes this condition is satisfied, and thus not discussed further.

the distributional sense, with first derivative:

$$\frac{\partial}{\partial K}(S_T - K)^+ = -\mathbb{I}_{\{K<S_T\}}$$

We now wish to compute the second derivative in K of $C(K,T)$. Recalling that we may interchange differentiation and integration, since the derivative of the payoff is dominated by an integrable function, we obtain:

$$\begin{aligned}\frac{\partial}{\partial K}C(K,T) &= e^{-rT}\int_{\mathbb{R}} \frac{\partial}{\partial K}(s-K)^+ f^{\mathbb{Q}}_{S_T}(s)\,ds \\ &= e^{-rT}\int_{\mathbb{R}} -\mathbb{I}_{\{K<s\}} f^{\mathbb{Q}}_{S_T}(s)\,ds \\ &= -e^{-rT}\int_K^{+\infty} f^{\mathbb{Q}}_{S_T}(s)\,ds \\ &= -e^{-rT}\,\mathbb{Q}(S_T > K)\end{aligned}$$

Recognizing that $\mathbb{Q}(S_T > K) = 1 - F^{\mathbb{Q}}_{S_T}(K)$, and that if $f^{\mathbb{Q}}_{S_T}$ exists then $F^{\mathbb{Q}}_{S_T}$ is differentiable, we can finally compute the second derivative:

$$\begin{aligned}\frac{\partial^2}{\partial K^2}C(K,T) &= -e^{-rT}\frac{\partial}{\partial K}\big(1 - F^{\mathbb{Q}}_{S_T}(K)\big) \\ &= e^{-rT} f^{\mathbb{Q}}_{S_T}(K)\end{aligned}$$

We have reached the Breeden–Litzenberger formula, which allows us to ***recover the risk–neutral distribution of the underlying asset using only the theoretical price of the call option***:

$$f^{\mathbb{Q}}_{S_T}(K) = e^{rT}\frac{\partial^2}{\partial K^2}C(K,T)$$

3. The "Real-World" Problem

The model developed so far is extremely interesting, yet it appears to be impossible to apply directly to the "real world", since the theoretical price $C(K,T)$ of a call option is not actually known. It is therefore necessary to bridge the gap that exists between theory and computational practice.

The underlying idea is relatively simple, but its correct implementation in Python requires some technical care. This paper does not aim to provide an exhaustive description of all implementation details of the algorithm; however, for readers interested in a deeper exploration, the script used is available in the following GitHub: .

As previously mentioned, we do not know the call option price surface, yet we can construct it. Figure 1 summarizes the full pipeline used to extract the risk-neutral information embedded in option prices. ***Starting from raw market quotes, we compute discrete implied volatilities, fit a smooth arbitrage-free volatility smile through SVI calibration, and use it to generate the continuous Call price surface via the Black–Scholes model. Finally, the second-order finite–difference approximation of the Call prices yields the risk-neutral density through the Breeden–Litzenberger formula.***

We now examine each of these steps in detail.

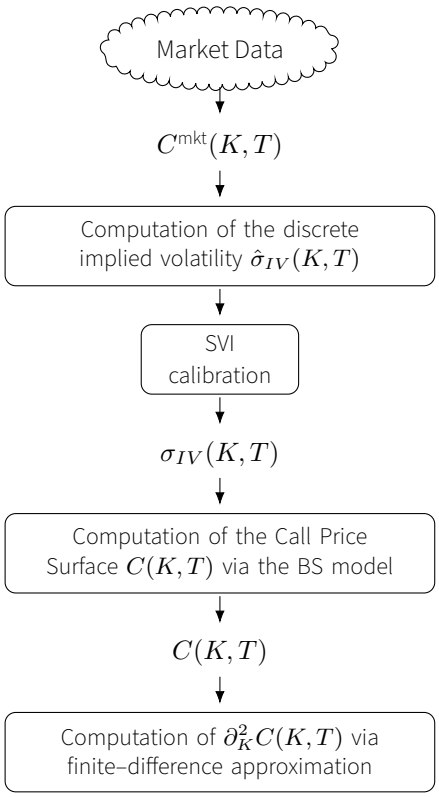

Figure 1: Workflow for obtaining $\partial^2_K C(K,T)$

3.1 Discrete Implied Volatility

The first step of the workflow consists in extracting the discrete implied volatility from option market quotes. Given the market price of a European Call option $C^{\text{mkt}}(K,T)$ for a set of strikes K_i and maturities T, the implied volatility $\hat{\sigma}_{IV}(K_i,T)$ is defined as the unique value of the volatility parameter σ that reproduces the observed price under the Black–Scholes model[2]:

$$C_{BS}(..., \hat{\sigma}_{IV}(K_i,T)) = C^{\text{mkt}}(K_i,T).$$

The outcome of this step is a collection of implied volatilities defined only at the strikes for which option prices are available in the market. This produces a ***discrete*** implied volatility smile.

3.2 SVI Calibration

A central step in the construction of a smooth and arbitrage-free implied volatility surface is the interpolation of the discrete implied volatilities $\hat{\sigma}_{IV}(K_i,T)$ observed in the market for each maturity T. Since market data naturally form a discrete and often noisy surface, a direct application of the Breeden–Litzenberger formula would be numerically unstable. To recover a reliable risk-neutral density, the function $\sigma_{IV}(K,T)$ must therefore be continuous, sufficiently regular, and defined for every strike K, so as to guarantee the existence and stability of its derivatives with respect to K.

To achieve this, we employ the Stochastic Volatility Inspired (SVI) parametrization, which should be regarded not merely as an interpolation method, but as a genuine model for the

[2] Since no closed-form solution for $\hat{\sigma}_{IV}$ exists, the inversion of the Black–Scholes formula requires a numerical root-finding procedure.

volatility smile. The SVI framework ensures the absence of butterfly arbitrage, produces a smooth and convex implied volatility curve, and enables robust extrapolation to strikes that are not quoted in the market.

The starting point is the notion of ***total implied variance***, defined for a fixed maturity T in terms of the true (and unknown) implied volatility $\sigma_{IV}(K,T)$ as

$$w(K,T) = \sigma_{IV}(K,T)^2\, T. \tag{2}$$

Since the shape of the implied volatility smile is more naturally described as a function of log-moneyness, we introduce

$$k = \log\left(\frac{K}{F_T}\right),$$

that is, the logarithmic distance of the strike from the forward F_T, which represents the risk–neutral expected price of the underlying and therefore the natural centre of its distribution. In other words, k measures how many times the strike is larger or smaller than the forward price, expressed on a logarithmic scale.[3]

The quantity we seek to model is therefore the total variance as a function of k:

$$w(k) := w(K,T) = \sigma_{IV}(K,T)^2\, T.$$

The SVI model postulates that the total variance function $w(k)$ can be represented as

$$w_{\text{SVI}}(k;\theta) = a + b\left(\rho(k-m) + \sqrt{(k-m)^2+\sigma^2}\right),$$

where $\theta = (a, b, \rho, m, \sigma)$.

Having clarified the theoretical foundations of the SVI calibration, we can now compute the market total variances w_i^{mkt} from the observed discrete implied volatilities $\hat{\sigma}_{IV}(K_i,T)$:

$$w_i^{\text{mkt}} = \hat{\sigma}_{IV}(K_i,T)^2\, T.$$

The calibration of the parameter vector θ is then carried out by minimizing the mean squared error between the total variances generated by the SVI model and the market-implied total variances, that is,

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{n}\sum_{i=1}^{n}\left(w_{\text{SVI}}(k_i;\theta) - w_i^{\text{mkt}}\right)^2.$$

$\hat{\theta}$ yields the best SVI approximation to the observed total variance smile. Once calibration is complete, the continuous implied volatility smile is recovered by inverting the equation (2):

$$\sigma_{IV}(K,T) = \sqrt{\frac{w_{\text{SVI}}(k;\hat{\theta})}{T}}$$

[3]The forward price F_T can be directly and reliably extracted from market data, so its computation will not be discussed further here. Curious readers are referred to the accompanying GitHub repository for implementation details.

3.3 Computation of $C(K,T)$ via Black–Scholes Model

At this point, we have moved from a discrete set of market observations to a smooth implied–volatility surface $\sigma_{IV}(K,T)$. However, in order to apply the Breeden–Litzenberger formula, we require the corresponding call price surface:

$$(K,T) \;\longmapsto\; C(K,T).$$

To obtain a continuous and arbitrage–free representation of $C(K,T)$, we reconstruct the call surface by evaluating the Black–Scholes formula at each point (K,T) using the implied volatility surface σ_{IV}.

Under the risk–neutral measure $\mathbb{Q}$, the underlying asset price follows a geometric Brownian motion,

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t,$$

The arbitrage–free price of a European call option with strike K and maturity T is given by the Black–Scholes formula

$$C^{\text{BS}}(S_0,K,T,r,\sigma) = S_0\,\mathcal{N}(d_1) - Ke^{-rT}\,\mathcal{N}(d_2), \tag{3}$$

with

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T},$$

and $\mathcal{N}(\cdot)$ denoting the standard normal cumulative distribution function.

Fix a maturity T. Substituting σ with σ_{IV} into (3) yields a regularized call price section:

$$C(\,\cdot\,,T) \;:=\; C^{\text{BS}}\big(S_0,K,T,r,\sigma_{IV}(K,T)\big),$$

that is, a function of the strike variable only. Collecting these sections for all maturities produces the full call price surface $C(K,T)$, which is continuous and differentiable in K, and therefore suitable for second–order differentiation in the Breeden–Litzenberger framework.

3.4 Finite–difference approximation

In order to extract the risk–neutral density from option prices, the Breeden–Litzenberger identity requires the computation of the second derivative of the call price with respect to the strike.

In our framework, the call price surface is obtained by evaluating the Black–Scholes formula at each point (K,T) using a fitted implied–volatility surface $\sigma_{IV}(K,T)$. Although the Black–Scholes pricing function is analytic in K for fixed volatility, the composition

$$K \;\longmapsto\; C(K,T) = C^{\text{BS}}\big(S_0,K,T,r,\sigma_{IV}(K,T)\big)$$

inherits a nontrivial dependence on K through $\sigma_{IV}(K,T)$, whose derivatives are not available in closed form. For practical purposes, it is therefore natural to approximate the second derivative $\partial_K^2 C(K,T)$ numerically on a strike grid by means of finite–difference methods.

Let $\{K_i\}_{i=1}^n$ denote an increasing and uniformly spaced grid of strikes, and let $C_i = C(K_i,T)$ be the corresponding call prices

for a fixed maturity T. For interior points ($i = 2, \ldots, n-1$) of the grid[4], the second derivative can be approximated by the standard central finite–difference scheme,

$$\frac{\partial^2 C}{\partial K^2}(K_i, T) \approx \frac{C_{i+1} - 2C_i + C_{i-1}}{(\Delta K)^2},$$

where $\Delta K = K_{i+1} - K_i$ denotes the strike spacing.

Once the second derivative $\partial_K^2 C(K,T)$ has been approximated on the strike grid, the Breeden–Litzenberger formula can be applied pointwise to obtain the numerical risk–neutral density. Since numerical differentiation and truncation may produce a function whose total mass differs from one, the Breeden–Litzenberger output does not automatically define a valid density.

For this reason, the approximated function is normalised by dividing it by its integral:

$$f_{S_T}^{\mathbb{Q}}(K) = \frac{f_{S_T}^{\text{num}}(K,T)}{\displaystyle\int f_{S_T}^{\text{num}}(u,T)\,du},$$

so that

$$\int f_{S_T}^{\mathbb{Q}}(K)\,dK = 1,$$

ensuring a proper probability density.

4. Model Validation on Synthetic Data

Before applying the full pipeline to real market data, it is essential to verify the correctness and numerical stability of the model in a fully controlled environment. To this end, we construct a synthetic dataset in which the volatility term structure, the option prices, and the corresponding risk–neutral density are analytically known. This allows us to quantitatively assess the reconstruction error at every step of the procedure.

To evaluate the accuracy of the recovered density, we compute three classical error norms:

$$\|f_{\text{num}} - f_{\text{true}}\|_{L^1}, \qquad \|f_{\text{num}} - f_{\text{true}}\|_{L^2}, \qquad \|f_{\text{num}} - f_{\text{true}}\|_{L^\infty},$$

which measure, respectively, the integral error, the mean–square error, and the maximum pointwise deviation between the numerical and the true density.

We further verify the consistency of the density normalization:

$$\int f_{\text{true}}(K,T)\,dK \approx 1, \qquad \int f_{\text{num}}(K,T)\,dK \approx 1,$$

a necessary condition for numerical stability. Significant deviations from unity would point to discretization issues in the second–order derivative with respect to the strike.

4.1 Results

It is worth reiterating that the purpose of this article is not to provide a full Python walkthrough. For this reason, we will not illustrate the implementation details of each computational step discussed in the previous sections; instead, we report directly the numerical results obtained.

As shown in Figure 2, the numerical density closely matches the theoretical one. This agreement is also reflected in the error values:

$$\|E\|_{L^1} \approx 1.3442 \times 10^{-4}; \qquad \|E\|_{L^2} \approx 5.1729 \times 10^{-6},$$
$$\|E\|_{L^\infty} \approx 1.4386 \times 10^{-6}; \qquad \int f_{\text{num}}(K,T)\,dK \approx 0.9891.$$

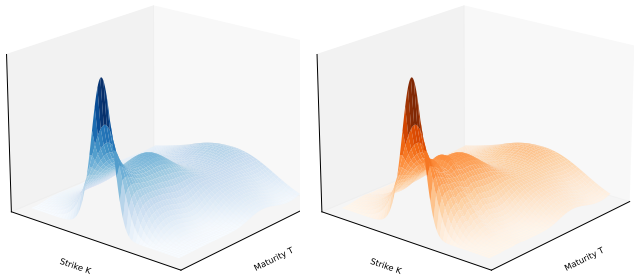

Figure 2: Comparison between the true density (left) and the numerically reconstructed density (right).

Overall, the synthetic–data experiment confirms that the proposed methodology is able to reconstruct a nontrivial risk–neutral density with high accuracy, thereby validating the approach before moving to real–market applications.

5. Results of the Application to the S&P 500

Having clarified both the theoretical framework and its practical implementation, we now present the empirical application of the model to the S&P 500 index.

After retrieving market option data, smoothing the discrete implied–volatility quotes into a continuous and arbitrage–free surface, reconstructing the corresponding call–price function, and evaluating its second derivative with respect to the strike, we obtain the Risk–Neutral Density associated with the future value of the index.

Figure 3 illustrates the resulting density surface. As expected, the distribution becomes increasingly concentrated around the current spot level for short maturities. Intuitively, when the time horizon is small, the underlying index has limited time to move significantly away from its present value, leading to a sharper, less dispersed density. For longer maturities, by contrast, the wider spread reflects the greater uncertainty that the market attributes to the long–term evolution of S_T.

Figure 4 complements this analysis by juxtaposing two pieces of information: (i) on the left, the historical price chart of the S&P 500, capturing the recent trajectory of the index; (ii) on the right, the heatmap representation of the inferred risk–neutral density $f_{S_T}^{\mathbb{Q}}$, where the color intensity encodes the probability mass assigned to each region of the (K,T) plane.

This joint visualization provides a richer and more coherent interpretation of market expectations. The historical chart

[4] At the boundaries of the grid, where symmetric differences cannot be formed, one–sided finite–difference formulas is employed.

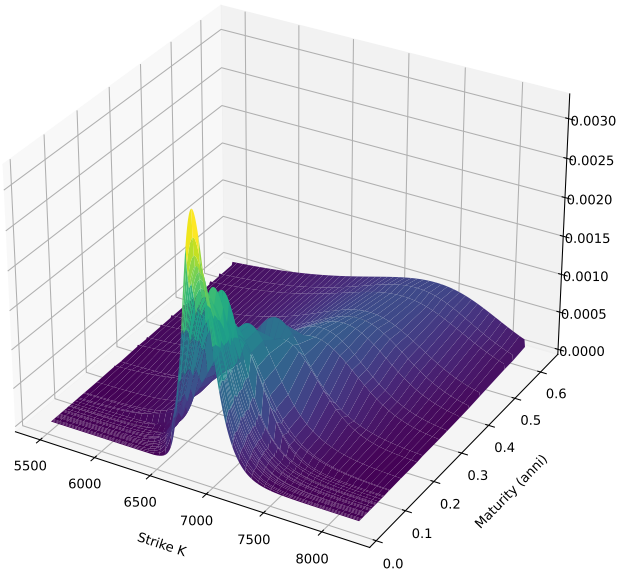

Figure 3: Risk–neutral density $f^{\mathbb{Q}}_{S_T}$ implied by S&P 500 option prices.

highlights the most recent trends (periods of appreciation, corrections, and volatility clusters) while the heatmap shows how such information is incorporated into option prices under the risk–neutral measure.

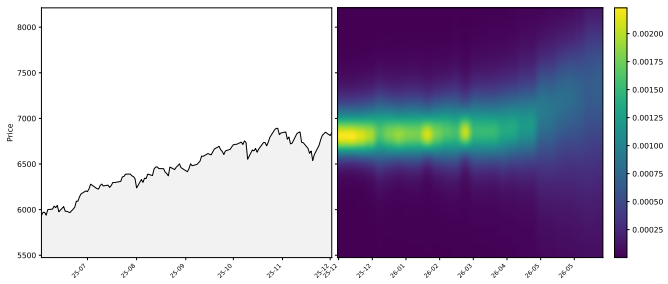

Figure 4: Historical price evolution of the S&P 500 (left) and heatmap of the implied risk–neutral density $f^{\mathbb{Q}}_{S_T}$ (right).

When the two visualizations are examined together, it becomes evident that the information contained in the implied–volatility surface (and consequently in the derived risk–neutral density) is broadly consistent with the realized dynamics of the index. Upward market phases tend to shift the probability mass towards higher strikes, while episodes of increased volatility produce a more dispersed and flatter density, reflecting greater uncertainty about future outcomes.

References

[1] S. E. Shreve, ***Stochastic Calculus for Finance II: Continuous-Time Models***. Springer, 2004.

[2] T. Björk, ***Arbitrage Theory in Continuous Time***, 3rd ed. Oxford University Press, 2009.

[3] M. Musiela and M. Rutkowski, ***Martingale Methods in Financial Modelling***. Springer, 2005.

[4] R. Rougé, ***Mathematical Foundations of Option Pricing***. Springer, 2023.

[5] D. T. Breeden and R. H. Litzenberger, "Prices of state-contingent claims implicit in option prices," ***Journal of Business***, vol. 51, no. 4, 1978.

[6] J. Gatheral, "A parsimonious arbitrage-free implied volatility parameterization," ***Global Derivatives Conference, Madrid***, 2004.

[7] A. Quarteroni, R. Sacco, F. Saleri, and P. Gervasio, ***Matematica Numerica*** (UNITEXT), 4th ed. Springer, 2019, ISBN: 978-88-470-1510-2.

[8] A. M. Malz, "A simple and reliable way to compute option-based risk-neutral distributions," ***FRBNY Staff Report***, no. 677, 2014.

[9] F. Black and M. S. Scholes, "The pricing of options and corporate liabilities," ***Journal of Political Economy***, vol. 81, no. 3, 1973.

[10] R. C. Merton, "Theory of rational option pricing," ***Bell Journal of Economics and Management Science***, vol. 4, 1973.

[11] M. Rossi, "Risk-neutral density recovery and option pricing," ***Journal of Banking & Finance***, vol. 72, 2016.

[12] M. R. Fengler, "Arbitrage-free smoothing of the implied volatility surface," ***Quantitative Finance***, vol. 9, no. 4, 2009.

[13] A. Itkin, "Pricing and calibration with smoothed implied volatility surfaces," ***Applied Mathematical Finance***, 2017.

[14] Python Software Foundation, ***Python: A programming language***, 2009.

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